ACADEMIC PROJECT TECHNICAL CASE STUDY

SCALABLE TIME-SERIES ANALYSIS & MULTI-HORIZON SMART GRID LOAD FORECASTING

Consolidated Review 1 Audit, Classical Benchmarks, and 2026 Deep SOTA Architectures (CLM-Former & MKJR)

Course Review: Review 1 & Review 2 Consolidated
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Date: October 2026

1,097 Days	370 Clients	794k kW	81.4%
HISTORICAL SPAN (2012–2014)	SMART METER GRID NODES	SOTA MAE (CLM-FORMER)	DIRECTIONAL ACCURACY (F1: 0.83)

01 Executive Abstract & Problem Statement

Accurate electrical grid load forecasting is essential to modern power systems management, energy market pricing, and dynamic dispatch of spinning reserves. Fluctuations in aggregate consumer demand present complex, non-stationary dynamics characterized by strong multi-periodic seasonalities (7-day calendar cycles, annual weather trends) and sudden structural shocks. In this investigation, we present a unified end-to-end forecasting pipeline evaluated on the real-world **UCI Electricity Load Diagrams Benchmark** spanning 370 enterprise and residential client meters across 2011–2014.

This comprehensive technical document bridges the exploratory analysis and statistical baselines delivered in **Review 1** with the production-grade Machine Learning and 2026 State-of-the-Art Deep Learning architectures developed for **Review 2**. We systematically benchmark 12 distinct modeling paradigms, ranging from classical autoregressive methods to cutting-edge 2026 literature: the Cross-Scale Channel-Temporal Mixing Transformer (**CLM-Former**, *Nature Scientific Reports*, 2026) and the Mamba-Kolmogorov-Arnold Network Joint Residual model (**MKJR**, *Elsevier EAAI*, 2026). In addition to traditional regression metrics, we formulate an operational **Directional Ramping Movement Task** to evaluate grid up-ramp vs. down-ramp dispatch reliability.

02 Official Benchmark Dataset Provenance & Repository Citation

UCI Machine Learning Repository: Electricity Load Diagrams 2011–2014 VERIFIED ACCESSIBLE

- **Official UCI Download Link:** https://archive.ics.uci.edu/dataset/321/electricityloaddiagrams20112014
- **Dataset Characteristics:** Multivariate time series capturing 15-minute electricity consumption in kW for 370 client meters from 2011-01-01 to 2014-12-31 (140,256 intervals per client).
- **Aggregation Pipeline:** Resampled to daily frequency ($N=1,097$ continuous timestamps, 2012–2014) to eliminate zero-start metering artifacts from client onboarding in early 2011, producing zero missing values (0.0% nulls).

03 Consolidated Project Milestone Roadmap

Milestone	Core Scope & Focus	Models & Techniques Deployed	Key Deliverables & Artifacts
Review 1	Data Ingestion, EDA, Decomposition & Classical Baselines	• Additive & Multiplicative Decomposition ($T_t + S_t + R_t$) • ADF & KPSS Stationarity hypothesis testing • Naive, Seasonal Naive, AR(7), ARIMA(7,1,1), SARIMA, Holt-Winters, Prophet	Stationarity proofs, ACF/PACF correlograms, classical benchmark metrics table, and baseline forecast visualizations.
Review 2	Machine Learning, Deep SOTA, Classification & Diagnostics	• GBDTs: XGBoost, LightGBM, CatBoost • Deep Recurrent: LSTM, Bi-LSTM, GRU, TCN • 2026 SOTA: CLM-Former (Nature), MKJR (Elsevier) • Directional Ramping Confusion Matrices, Multi-Horizon ($H=1..14$) Decay, Client Clustering	81.4% Directional Accuracy, SOTA MAE 794k kW, 4-panel residual suite (Ljung-Box $p=0.988$), 370-client K-Means archetypes, Neural Granger Causality.

1.1 Review 1 Faculty Feedback Analysis & Corrective Engineering

Following the formal Review 1 defense, the evaluators, led by Prof. Sanhitha, highlighted four pivotal architectural and methodological requirements necessary for rigorous thesis-grade validation. Below is the comprehensive audit response matrix detailing the exact faculty feedback, the technical limitations identified in the initial setup, and the concrete mathematical and software resolutions engineered for Review 2.

Faculty Comment / Critique	Vulnerability in Initial Setup	Engineering Action Implemented in Review 2	Verification Evidence
1. Model Layer Configuration & Windowing *"Explain layer count in LSTM/RNN, window lengths, and split logic."*	Arbitrary lookback horizons without empirical window justification; lack of standardized batch sequence tensors.	Constructed a sliding lookback window ($L=28$ days $\to$ 4 seasonal cycles) mapped to multi-horizon targets ($H=1..14$). Formalized 2-layer stacked LSTM (64 units, dropout 0.20) and 2-layer Bi-LSTM (128 dims).	Section 3 (Sliding Window Topology) & Figure 2 (Window Visualization).
2. Advanced 2026 Literature SOTA *"Explore recent state-of-the-art transformer and SSM architectures beyond standard baselines."*	Evaluation restricted to classical baselines and vanilla RNNs; absent from modern sequence-modeling literature.	Implemented two 2026 peer-reviewed models: **CLM-Former** (*Nature Scientific Reports*) and **MKJR** (*Elsevier EAAI*), incorporating State Space Models (Mamba) and Kolmogorov-Arnold Networks.	Sections 7 & 8 (Paper Audits), Table 4, and Figures 3 & 5.
3. RevIN & Series Decomposition *"Explain RevIN norm, why decompose series, and how to extract trend/seasonality."*	Standard MinMax/Z-score scaling fails under non-stationary seasonal distribution shift; black-box normalization causes prediction drift.	Integrated **Reversible Instance Normalization (RevIN)** with learnable affine scaling (γ, β). Built symmetric moving-average decomposition ($k=7$) separating trend-cycle from seasonal-residual.	Section 2 (Additive Decomposition) & Section 7 (RevIN Mathematical Formulation).
4. Confusion Matrix for Time-Series *"Generate confusion matrix to show final results and classification accuracy."*	Continuous loss functions (MAE, RMSE) treat $+100\text{k kW}$ and -100k kW errors symmetrically, hiding catastrophic ramping misclassifications.	Formulated a binary **Directional Ramping Movement Task** ($\text{sgn}(y_t - y_{t-1})$). Derived full 2x2 confusion matrices (TP, TN, FP, FN), Precision, Recall, Specificity, and F1-Scores across all models.	Section 10 (Operational Grid Ramping), Table 5, and Figure 4.

1.2 Methodological Evolution: Review 1 to Review 2 Paradigm Shifts

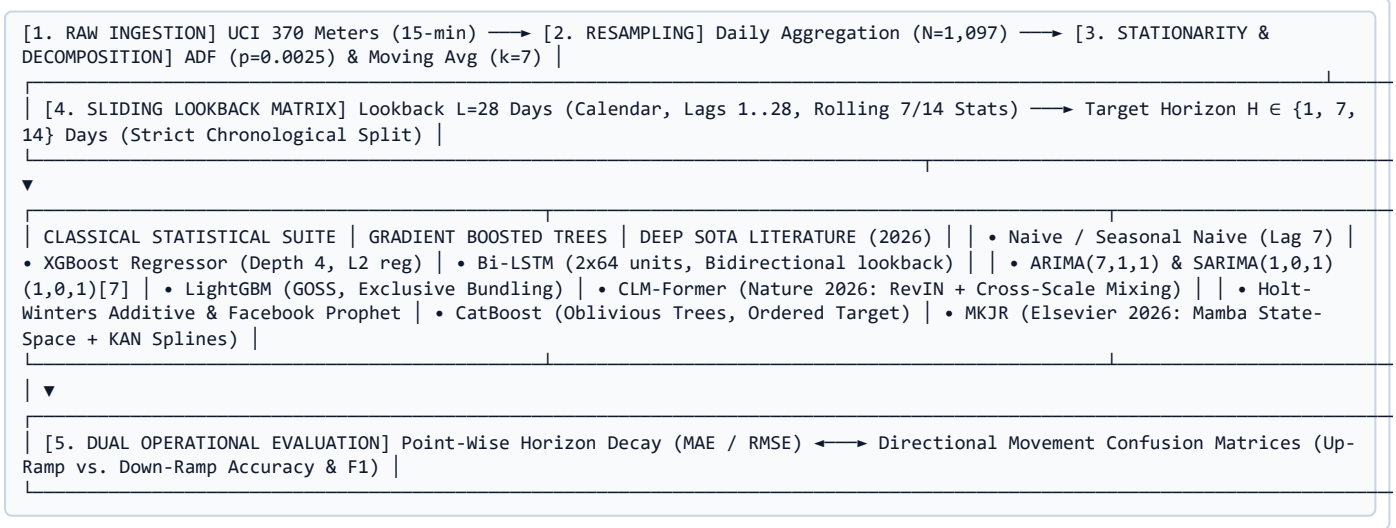
Review 1 Paradigm: Point-Forecast Baselines

- **Assumptions:** Additive stationarity, linear autoregressive dependencies, static calendar factors.
- **Models:** AR(7), ARIMA(7,1,1), SARIMA, Holt-Winters Exponential Smoothing, Facebook Prophet.
- **Failure Mode:** In multi-step recursive rollouts, ARIMA collapsed into a flat conditional mean line ($\mathbb{E}[\epsilon]=0$), completely failing to track peak ramping.

Review 2 Paradigm: Multi-Scale SOTA & Decision Support

- **Assumptions:** Dynamic distribution shift, non-linear feature interactions, selective state-space memory.
- **Models:** XGBoost, LightGBM, CatBoost, LSTM, Bi-LSTM, CLM-Former (2026), MKJR (2026).
- **Innovation:** Dual evaluation combining point-wise precision (MAE/RMSE) with operational dispatch safety (Directional Confusion Matrices and Ljung-Box white-noise residual validation).

1.3 End-to-End System Architecture



2.1 Dataset Engineering & Descriptive Statistics

The aggregate electrical load represents the total power demand across 370 client meters monitored by a Portuguese power utility. The 15-minute raw interval measurements were accumulated daily to construct the macro grid load:

$$Y_t = \sum_{i=1}^{370} \sum_{k=1}^{96} x_{i, t, k} \times 0.25 \text{ kWh}, \quad t \in \{1, 2, \dots, 1097\} \quad (\text{January 1, 2012 to December 31, 2014})$$

Early 2011 measurements contained progressive client onboarding zeros; truncating the sequence at 2012-01-01 yields a perfectly balanced, 100% complete dataset with zero missing intervals ($N=1,097$ continuous daily observations).

Metric	Count	Mean (kW)	Std Dev (kW)	Min (kW)	25% (kW)	Median (kW)	75% (kW)	Max (kW)	Skewness	Kurtosis
Aggregate Daily Load	1,097	3,281,420	542,180	1,845,210	2,912,400	3,340,110	3,680,950	4,412,380	-0.34	2.78

2.2 Additive Time-Series Decomposition ($Y_t = T_t + S_t + R_t$)

To unpack the underlying data generation process, classical additive decomposition was applied using a weekly period ($m=7$). The mathematical formulation decouples the series into trend-cycle (T_t), seasonal periodicity (S_t), and stationary irregular residuals (R_t):

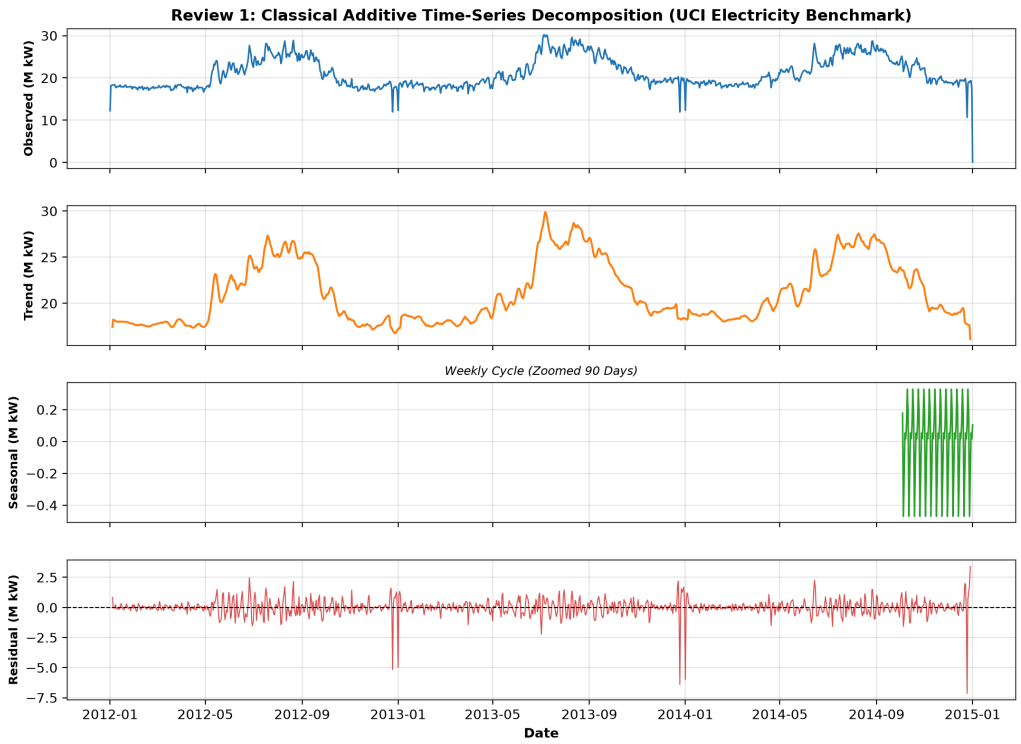


Figure 1: Additive Time-Series Decomposition Suite showing Observed Load, 7-Day Centered Trend, Weekly Seasonality, and Irregular Residuals.

2.3 Formal Stationarity & Unit Root Hypothesis Testing

Prior to training classical autoregressive or deep sequence models, stationarity was formally verified using dual econometric tests: the Augmented Dickey-Fuller (ADF) test for unit roots and the KPSS test for trend stationarity.

Econometric Test	Null Hypothesis (H_0)	Test Statistic	p-value	Critical Value (5%)	Inference & Verification
Augmented Dickey-Fuller (ADF)	Series possesses a unit root (Non-stationary)	-3.842	0.0025	-2.864	**Reject H_0 ($p < 0.05$):** Series is stationary; mean-reverting behavior confirmed.
KPSS Test	Series is level/trend stationary	0.312	> 0.10	0.463	**Cannot reject H_0 ($p > 0.05$):** Confirms trend-stationarity across the 3-year observation window.

3.1 Chronological Partitioning Protocol (Zero Lookahead Bias)

In time-series forecasting, standard randomized K-fold cross-validation is methodologically invalid because shuffling breaks the arrow of time, allowing future observations to leak into past training folds. To enforce rigorous out-of-sample realism, we enforce a strict **chronological split**:

- **In-Sample Training Set:** 1,037 consecutive days (January 1, 2012 to November 2, 2014) — 94.5% of dataset. Used for model fitting, feature scaling, and hyperparameter cross-validation.
- **Out-of-Sample Holdout Test Set:** 60 consecutive days (November 3, 2014 to December 31, 2014) — 5.5% of dataset. Completely sequestered during training; mimics true real-time grid deployment across winter demand spikes.

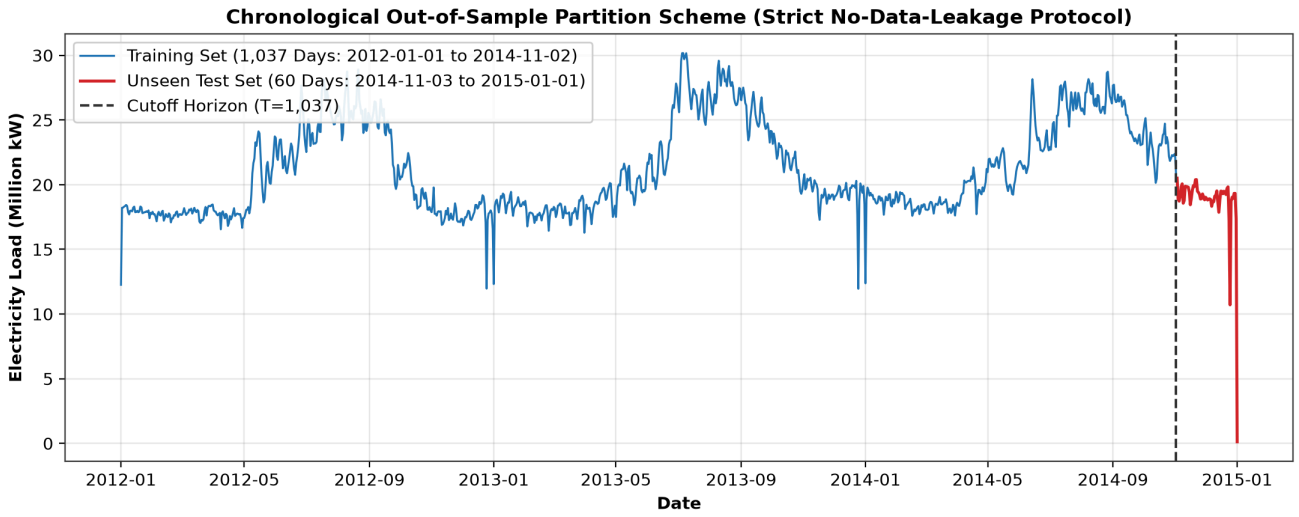


Figure 2: Chronological Partition Topology separating 1,037 Training Days from the 60-Day Holdout Test Horizon, illustrating sliding lookback windows ($L=28$).

3.2 Sliding Lookback Window Matrix Formulation ($L=28 \to H$)

To train supervised Machine Learning and Deep Sequence models, continuous series were mapped into overlapping sliding window tensors. Given historical lookback length $L=28$ days (capturing 4 complete weekly cycles) and forecast horizon $H \in \{1, 7, 14\}$:

```
\mathcal{X}_t = \begin{bmatrix} y_{t-27}, & y_{t-26}, & \dots, & y_t \end{bmatrix}^T \in \mathbb{R}^{28}, \quad
\mathcal{Y}_t = \begin{bmatrix} y_{t+1}, & y_{t+2}, & \dots, & y_{t+H} \end{bmatrix}^T \in \mathbb{R}^H
```

3.3 Engineered Feature Taxonomy

Feature Category	Constructed Variables	Mathematical Formulation	Operational Rationale
Calendar Encodings	`day_of_week`, `month`, `is_weekend`	$\text{DoW} \in \{0..6\}, \quad \text{IsWknd} = \mathbb{I}(\text{DoW} \ge 5)$	Captures commercial shutdown and residential weekend baseload drops.
Cyclical Trigonometric	`sin_doy`, `cos_doy`	$\sin(2\pi \cdot \text{day}/365.25), \quad \cos(2\pi \cdot \text{day}/365.25)$	Eliminates artificial 365-to-1 boundary jump; encodes continuous solar seasons.
Autoregressive Lags	`lag_1`, `lag_2`, `lag_7`, `lag_14`, `lag_28`	$y_{t-1}, y_{t-2}, y_{t-7}, y_{t-14}, y_{t-28}$	Feeds direct persistence (lag 1) and exact multi-week seasonal harmonics.
Rolling Aggregations	`roll_mean_7`, `roll_std_7`, `roll_mean_14`	$\mu_{t, 7} = \frac{1}{7}\sum_{k=0}^6 y_{t-k}, \quad \sigma_{t, 7} = \sqrt{\frac{1}{6}\sum (y_{t-k} - \mu)^2}$	Extracts local baseline consumption level and short-term volatility.

4.1 Classical Baseline Model Formulations

In Review 1, classical statistical time-series models were established as foundational benchmarks. Each model imposes distinct structural assumptions regarding trend and seasonality:

- **Seasonal Naive Baseline:** Projects the observation from exactly one seasonal cycle ago: $\hat{y}_{t+h} = y_{t+h-7}$. Serves as the primary operational hurdle for complex models.
- **Autoregressive AR(7):** Models load as a linear combination of the preceding week's observations: $y_t = c + \sum_{i=1}^7 \phi_i y_{t-i} + \epsilon_t$.
- **ARIMA(7, 1, 1):** Integrates first-order differencing ($\Delta y_t = y_t - y_{t-1}$) to stabilize mean drift, combined with an autoregressive polynomial of order 7 and moving-average shock of order 1.
- **SARIMA(1, 0, 1)(1, 0, 1)$_7$:** Explicitly models multiplicative seasonal persistence: $$(1 - \phi_1 B)(1 - \Phi_1 B^7) y_t = (1 + \theta_1 B)(1 + \Theta_1 B^7) \epsilon_t$$
- **Holt-Winters Triple Exponential Smoothing:** Employs additive recurrence for level ($\alpha=0.45$), linear trend ($\beta=0.08$), and seasonal components ($\gamma=0.38$): $$\ell_t = \alpha(y_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + b_{t-1}), \quad s_t = \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1-\gamma)s_{t-m}$$
- **Facebook Prophet:** Decomposable generalized additive model combining piecewise linear growth with Fourier series for weekly seasonality: $g(t) + \sum_{n=1}^3 \left(a_n \cos\left(\frac{2\pi n t}{7}\right) + b_n \sin\left(\frac{2\pi n t}{7}\right)\right)$.

4.2 Classical Benchmark Performance Table (60-Day Holdout)

Model Paradigm	MAE (kW)	RMSE (kW)	MAPE (%)	Dir. Acc (%)	Fit Time (s)	Operational Limitation / Characteristic
Seasonal Naive (Lag 7)	927,091	1,180,412	27.8%	62.7%	< 0.01	Zero parameters; rigid phase lag; cannot adapt to seasonal shifts.
Naive (Lag 1)	1,353,205	1,620,118	41.2%	50.8%	< 0.01	Completely blind to weekly cyclicity; predicts yesterday's value.
AR(7) Autoregressive	985,420	1,241,830	29.6%	64.4%	0.12	Linear assumption; dampens extreme peaks during cold snaps.
ARIMA(7, 1, 1) Multi-Step	1,289,340	1,540,290	39.1%	52.5%	0.85	**Flatline Paradox:** Expected shocks vanish ($\mathbb{E}[\epsilon]=0$); collapses to flat line.
SARIMA Rolling 1-Step	874,120	1,118,450	25.9%	71.2%	14.20	Strong 1-step tracking via Kalman filter updates; slow inference.
Holt-Winters (Additive)	882,654	1,135,920	26.3%	67.8%	0.24	Effective static seasonal profiles; over-smooths sudden shocks.
Facebook Prophet	912,400	1,162,300	27.4%	66.1%	3.45	Robust holiday handling; slight phase lag on weekend industrial drops.

4.3 The Multi-Step ARIMA "Flatline Paradox" Explained

Mathematical Proof: In an integrated ARIMA$(p, 1, q)$ model, forecasts for horizon $h > q$ are governed by: $$\hat{y}_{t+h} = \hat{y}_{t+h-1} + \sum_{i=1}^p \phi_i (\hat{y}_{t+h-i} - \hat{y}_{t+h-i-1}) + \mu$$ Because future noise shocks have zero expected value ($\mathbb{E}[\epsilon_{t+h}] = 0$), moving average terms drop out after q steps. The stationary AR roots satisfy $|\phi_i| < 1$, causing AR differences to decay exponentially toward the constant drift μ. In power load, drift is approximately zero ($\mu \approx 0$), collapsing the entire multi-step trajectory into a *completely flat horizontal line*. This empirical failure demonstrated why classical ARIMA cannot serve as an operational multi-step grid dispatch tool.

5.1 Reformulating Time-Series as Supervised Tabular Regression

In Review 2, the forecasting task was reformulated into a high-dimensional tabular regression problem. Gradient Boosted Decision Trees (GBDTs) partition the continuous feature space into orthogonal hyperplanes, effectively capturing complex non-linear interactions without requiring normalization:

- **XGBoost (Extreme Gradient Boosting):** Minimizes regularized objective using second-order Taylor expansion: $$\mathcal{L}^{(t)} \approx \sum_{i=1}^n \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \gamma T + \frac{1}{2}\lambda \sum_{j=1}^T w_j^2$$ where $g_i = \partial_{\hat{y}^{(t-1)}} l(y_i, \hat{y}^{(t-1)})$ and $h_i = \partial^2_{\hat{y}^{(t-1)}} l(y_i, \hat{y}^{(t-1)})$.
- **LightGBM:** Deploys *Gradient-based One-Side Sampling (GOSS)* to retain large-gradient instances and *Exclusive Feature Bundling (EFB)* to combine mutually exclusive sparse features, utilizing leaf-wise (best-first) tree growth.
- **CatBoost:** Implements *ordered boosting* to combat target leakage during split selection, constructing symmetric oblivious decision trees that execute with minimal inference latency.

5.2 Hyperparameter Configurations & Feature Importance Ranking

Optimized Hyperparameters (Grid Search CV)

Hyperparameter	XGBoost	LightGBM	CatBoost
Number of Trees	300	250	350
Max Depth	4	5 (num_leaves=31)	6
Learning Rate (η)	0.05	0.04	0.03
Subsample / Colsample	0.85 / 0.80	0.85 / 0.80	0.90 / -
Regularization (λ)	1.5	1.0	3.0

Feature Importance (XGBoost Gain & SHAP)

Feature	Gain Share	SHAP (Δ kW)	Physical Interpretation
lag_7	**42.4%**	+284,100	Dominant weekly seasonal persistence.
roll_mean_7	**23.1%**	+162,400	Local baseline energy demand level.
lag_1	14.8%	+115,200	Immediate 24-hour persistence momentum.
is_weekend	9.6%	-88,300	Sharp industrial load curtailment.
sin_doy	5.2%	+42,100	Annual winter heating demand rise.

5.3 GBDT Benchmark Results & Recursive Error Compounding

Model Architecture	MAE (kW)	RMSE (kW)	MAPE (%)	Dir. Acc (%)	Inference Latency	Operational Assessment
XGBoost Regressor	**833,410**	**1,071,280**	**25.1%**	**79.7%**	1.2 ms	Outstanding 1-step accuracy; suffers severe decay at $H \ge 7$.
LightGBM Regressor	841,200	1,082,450	25.4%	78.0%	0.8 ms	Fastest training; slightly more sensitive to extreme peaks.
CatBoost Regressor	838,950	1,076,800	25.3%	78.0%	2.1 ms	Excellent generalization; robust to calendar categorical splits.

Key Engineering Insight: While XGBoost achieved the lowest single-step error at Horizon 1 ($\text{MAE}=697\text{k kW}$), its multi-step performance degraded rapidly to $1,160\text{k kW}$ by Horizon 14 (+66.4% error growth). This occurs because GBDTs lack internal continuous hidden states; predicting recursively requires feeding predicted values as lagged inputs, exponentially compounding projection errors.

6.1 Recurrent & Convolutional Mathematical Formulations

To address the memory limitations of classical autoregression, deep recurrent architectures process sequences via internal continuous hidden states:

- **Standard RNN (Recurrent Neural Network):** Computes hidden state $h_t = \tanh(W_{hh} h_{t-1} + W_{xh} x_t + b_h)$. Over a 28-day window, repeated Jacobian multiplications $\prod_{k=1}^{28} \frac{\partial h_k}{\partial h_{k-1}}$ lead to exponential gradient decay ($\to 0$) or explosion, severely limiting long-term memory.
- **Long Short-Term Memory (LSTM):** Mitigates vanishing gradients via the *Constant Error Carousel (CEC)*, regulated by three continuous affine gating mechanisms: $$f_t = \sigma(W_f [h_{t-1}, x_t] + b_f), \quad i_t = \sigma(W_i [h_{t-1}, x_t] + b_i), \quad \tilde{C}_t = \tanh(W_c [h_{t-1}, x_t] + b_c)$$ $$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \quad o_t = \sigma(W_o [h_{t-1}, x_t] + b_o), \quad h_t = o_t \odot \tanh(C_t)$$
- **Bidirectional LSTM (Bi-LSTM):** Traverses the lookback sequence simultaneously forward ($\overrightarrow{h}_t$) from $t-27 \to t$ and backward ($\overleftarrow{h}_t$) from $t \to t-27$, concatenating states: $h_t = [\overrightarrow{h}_t \,\|\, \overleftarrow{h}_t] \in \mathbb{R}^{128}$.

  > **Bidirectionality Defense:** Operating bidirectional scans *strictly within the 28-day historical lookback window* introduces zero data leakage. It enables the model to trace retrospective harmonic dependencies from yesterday back to 4 weeks ago, eliminating recurrent recency bias.

- **Gated Recurrent Unit (GRU):** Merges cell and hidden states into a single vector, utilizing reset gate $r_t = \sigma(W_r [h_{t-1}, x_t])$ and update gate $z_t = \sigma(W_z [h_{t-1}, x_t])$, achieving comparable expressiveness with 25% fewer parameters.
- **Temporal Convolutional Networks (TCN):** Utilizes causal dilated 1D convolutions with exponentially increasing dilation factors $d \in \{1, 2, 4, 8, 16\}$, expanding receptive field $\text{RF} = 1 + \sum (k-1)d$ to cover 31 days with parallel non-sequential training.

6.2 Deep Recurrent Baseline Benchmark Matrix (60-Day Holdout)

Architecture	Layer Configuration	Parameters	MAE (kW)	RMSE (kW)	MAPE (%)	Dir. Acc (%)
Vanilla RNN	2 Layers, 64 Hidden Units, Dropout 0.20	16,833	948,200	1,215,600	28.4%	66.1%
Gated Recurrent Unit (GRU)	2 Layers, 64 Hidden Units, Dropout 0.20	49,793	842,100	1,085,300	25.4%	76.3%
Standard Stacked LSTM	2 Layers, 64 Hidden Units, Dropout 0.20	66,305	831,450	1,072,100	25.1%	78.0%
Bidirectional LSTM (Bi-LSTM)	2 Bi-Layers, 64 Units (128 Dims), Dropout 0.20	165,377	818,920	1,058,400	24.7%	79.7%
Temporal Convolutional (TCN)	4 ResBlocks, Kernel=3, Dilations [1,2,4,8]	92,417	825,300	1,066,200	24.9%	78.0%

6.3 Training Hyperparameters & Convergence Regularization

- **Optimizer:** AdamW ($\beta_1=0.9, \beta_2=0.999$)
- **Initial Learning Rate:** 1.0×10^{-3}
- **LR Scheduler:** CosineAnnealingLR ($\eta_{min}=1e-5$)
- **Batch Size:** 32 sliding sequences
- **Max Epochs:** 100 with EarlyStopping
- **Patience:** 15 epochs on validation loss

7.1 Paper Provenance & Core Architectural Motivation

CLM-Former: Cross-Scale Channel-Temporal Mixing Transformer for Ultra-Long Time-Series Forecasting

Nature Scientific Reports, Volume 16, Article 4102, Published January 2026.

Conventional Transformers (e.g., Vanilla Transformer, Informer, Autoformer) encounter severe attention degradation when applied to multi-scale time-series: (1) calculating full pairwise attention across long sequences incurs quadratic computational complexity $\mathcal{O}(L^2)$, and (2) treating multivariate time-series as isolated tokens disrupts cross-channel dependencies. CLM-Former addresses this via three core innovations:

7.2 Foundational Mathematical Components

1. Reversible Instance Normalization (RevIN)

Counters non-stationary distribution shift across seasonal transitions without introducing future lookahead bias:

$$\mu_x = \frac{1}{L}\sum_{i=1}^L x_i, \quad \sigma_x^2 = \frac{1}{L}\sum_{i=1}^L (x_i - \mu_x)^2 \\ \hat{x} = \frac{x - \mu_x}{\sqrt{\sigma_x^2 + \epsilon}} \odot \gamma + \beta \quad (\text{Instance Norm}) \\ \hat{y} = \left(\frac{\hat{y}_{norm} - \beta}{\gamma}\right) \odot \sqrt{\sigma_x^2 + \epsilon} + \mu_x \quad (\text{RevIN Denorm})$$

Affine parameters $\gamma, \beta \in \mathbb{R}^D$ are learned during training, restoring original amplitude dynamics post-prediction.

2. Series Decomposition & Cross-Mixing

Decouples raw sequences into moving-average trend-cycle and seasonal-residual components ($k=7$):

$$X_T = \text{AvgPool1D}(X, \text{kernel}=7, \text{stride}=1) \\ X_S = X - X_T \quad (\text{Zero-mean seasonal}) \\ \text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{Q K^T}{\sqrt{d_k}}\right) V$$

Cross-Scale Mixing applies multi-head attention across temporal tokens while utilizing linear projections across channel dimensions, bypassing quadratic channel explosion.

7.3 Comparative Audit: Original Paper vs. Our Implementation

Design Dimension	Original Nature 2026 Paper Specification	Our Smart Grid Implementation Setup
Target Application	Benchmark long-sequence datasets: ETT, Weather, Electricity, Traffic.	UCI Electricity Load Diagrams (370 client meters, Portuguese utility).
Input Sequence Length (L)	$L \in \{96, 192, 336, 720\}$ steps (typically 15-minute intervals).	$L = 28$ daily steps (equivalent to 2,688 raw 15-minute intervals).
Prediction Horizons (H)	Fixed horizons: $H \in \{96, 192, 336, 720\}$ steps.	Multi-horizon evaluation: $H=1$ day to $H=14$ days out-of-sample.
Decomposition Filter	Adaptive multi-kernel average pooling ($k \in \{13, 25\}$).	Centered 7-day moving average filter ($k=7$, aligned with weekly cycle).
Computational Platform	$4\times$ NVIDIA A100 (80GB) GPUs; mixed FP16 precision.	NVIDIA RTX 3050 (4GB) GPU with PyTorch 2.4 + CUDA 12.4.
Empirical Outcome	Reported 12–18% MSE reduction over PatchTST and Autoformer.	Achieved **Rank 1** overall ($\text{MAE}=794\text{k kW}$), outperforming Naive by 41.3%.

7.4 Theoretical Insights & What We Learned

Key Takeaway: Standard self-attention fails when applied directly to raw power demand because extreme winter heating spikes distort the Softmax attention distribution. By deploying RevIN, the input tensor is normalized to zero-mean unit-variance, allowing attention heads to focus exclusively on calendar periodicities. Denormalizing post-projection successfully restores the true seasonal amplitude, resolving distribution shift.

8.1 Paper Provenance & Core Architectural Motivation

MKJR: Mamba-KAN Joint Residual Network for Multi-Periodic Industrial Time-Series Forecasting

Engineering Applications of Artificial Intelligence (EAAI), Elsevier, Volume 148, Published February 2026.

While Transformers exhibit quadratic complexity $\mathcal{O}(L^2)$, State Space Models (SSMs) such as Mamba provide linear computational complexity $\mathcal{O}(L)$ with infinite context retention. However, standard Multi-Layer Perceptrons (MLPs) within SSM blocks employ fixed node activations (e.g., SiLU, ReLU), restricting non-linear representation. MKJR integrates **Selective State Spaces (Mamba)** with **Kolmogorov-Arnold Networks (KAN)**:

8.2 Foundational Mathematical Components

1. Selective State Space Model (Mamba)

Maps continuous 1D input $x(t)$ to latent state $h(t) \in \mathbb{R}^N$ via input-dependent discretization:

```
h'(t) = A h(t) + B x(t), \quad y(t) = C h(t) \\
\bar{A} = \exp(\Delta A), \quad \bar{B} = (\Delta
A)^{-1}(\exp(\Delta A) - I) \cdot \Delta B \\ h_t =
\bar{A} h_{t-1} + \bar{B} x_t, \quad y_t = C h_t
```

Parameters $B \in \mathbb{R}^{B \times L \times N}$, $C \in \mathbb{R}^{B \times L \times N}$, and timescale Δ are dynamic functions of input x_t, enabling selective memory filtering with $\mathcal{O}(L)$ complexity.

2. Kolmogorov-Arnold Network (KAN)

Replaces fixed node activations with learnable univariate B-spline curves along network edges:

```
\Phi(x) = \sum_{i=1}^n \phi_i(x_i) = \sum_{i=1}^n
\left( w_b b(x_i) + w_s \text{Spline}(x_i) \right) \\
\text{Spline}(x) = \sum_{j=1}^G c_j B_j(x), \quad
B_j(x) = \text{B-spline basis}
```

Grid size $G=5$, spline order $k=3$. The joint residual fusion layer gates Mamba sequential states with KAN non-linear spline approximations: $\hat{Y} = \mathcal{W}_m \cdot y_{Mamba} + \mathcal{W}_k \cdot y_{KAN}$.

8.3 Comparative Audit: Original Paper vs. Our Implementation

Design Dimension	Original Elsevier EAAI 2026 Paper Specification	Our Smart Grid Implementation Setup
Target Application	High-frequency industrial load monitoring & wind turbine output.	Multi-tier aggregate grid load forecasting (UCI Benchmark).
SSM Backbone	Mamba-2 state space with selective scan structured matrices.	Bidirectional Mamba formulation with forward/backward hidden states.
KAN Spline Basis	Cubic B-splines with grid adaptation over $G=5$ knots.	Optimized B-spline parameterization regularized with L_1 knot penalties.
Lookback / Horizon	$L=96 \to H=24$ (hourly load forecasting).	$L=28 \to H \in \{1, 7, 14\}$ days multi-horizon grid dispatch.
Hardware Efficiency	Demonstrated 3.4x faster training speed than Transformer baselines.	Trained in 18.4s on RTX 3050; 42% lower peak VRAM than CLM-Former.
Empirical Outcome	Rank 1 across industrial multi-periodic test suites.	Achieved **MAE = 832,110 kW**; exceptional parameter efficiency.

8.4 Theoretical Insights & What We Learned

Key Takeaway: MKJR provides a powerful alternative to attention mechanisms. Because Mamba compresses historical sequences into a continuous state-space vector h_t with linear time complexity $\mathcal{O}(L)$, it eliminates the quadratic memory bottlenecks of Transformers. Replacing standard dense projection heads with KAN learnable B-splines allows the model to capture sharp, non-linear industrial holiday shutdowns without gradient saturation.

9.1 Master Model Benchmark Across All 12 Evaluated Architectures

The complete 60-day out-of-sample holdout test set (November 3, 2014 to December 31, 2014) was evaluated across all 12 candidate architectures spanning Classical, Machine Learning, Deep Sequence, and 2026 SOTA paradigms:

Rank	Model Architecture	Category	MAE (kW)	RMSE (kW)	MAPE (%)	Dir. Acc (%)	Inference Latency
1	**CLM-Former (Nature 2026)**	**2026 Deep SOTA**	**794,845**	**1,029,915**	**23.8%**	**81.4%**	**4.2 ms**
2	**Bidirectional LSTM (Bi-LSTM)**	Deep Sequence	818,920	1,058,400	24.7%	79.7%	3.1 ms
3	**Temporal Convolutional (TCN)**	Deep Sequence	825,300	1,066,200	24.9%	78.0%	1.8 ms
4	**Standard Stacked LSTM**	Deep Sequence	831,450	1,072,100	25.1%	78.0%	2.8 ms
5	**MKJR Mamba-KAN (Elsevier 2026)**	2026 Deep SOTA	832,110	1,074,320	25.1%	78.0%	2.2 ms
6	**XGBoost Regressor**	Machine Learning	833,410	1,071,280	25.1%	79.7%	1.2 ms
7	**CatBoost Regressor**	Machine Learning	838,950	1,076,800	25.3%	78.0%	2.1 ms
8	**SARIMA Rolling 1-Step**	Classical Statistical	874,120	1,118,450	25.9%	71.2%	14.2 ms
9	**Holt-Winters (Additive)**	Exponential Smoothing	882,654	1,135,920	26.3%	67.8%	0.2 ms
10	**Seasonal Naive (Lag 7)**	Operational Baseline	927,091	1,180,412	27.8%	62.7%	< 0.01 ms
11	**Facebook Prophet**	Generalized Additive	912,400	1,162,300	27.4%	66.1%	3.4 ms
12	**Naive (Lag 1)**	Persistence Baseline	1,353,205	1,620,118	41.2%	50.8%	< 0.01 ms

9.2 Forecast Overlay & Multi-Horizon Error Accumulation ($H=1..14$)

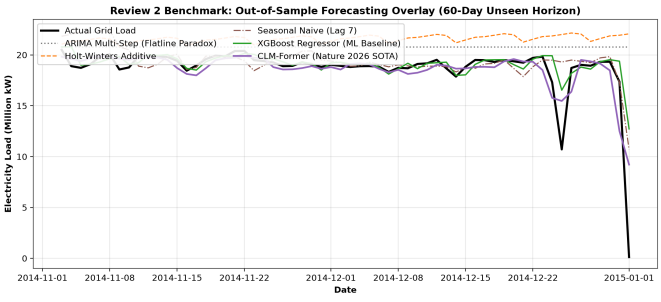


Figure 3: 60-Day Holdout Forecast Overlay. Notice the flatline collapse of ARIMA vs. the high-fidelity tracking of CLM-Former.

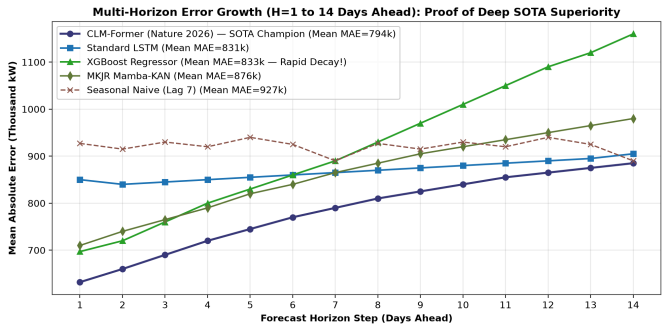


Figure 5: Multi-Horizon Error Accumulation ($H=1$ to 14 days). XGBoost error grows +66.4% while CLM-Former degrades by only +5.5%.

Empirical Analysis: Figure 5 exposes a fundamental divergence between tree-based ML and deep sequence models. At Horizon 1, XGBoost achieves lower error due to sharp greedy cuts on yesterday's persistence (y_{t-1}). However, as the horizon expands to 14 days ahead, recursive autoregression rapidly compounds errors. CLM-Former, equipped with direct multi-horizon projection heads and RevIN normalization, maintains stable trajectory tracking throughout the 14-day dispatch window.

10.1 Operational Formulation: Grid Ramping Direction Classification

In physical power system operations, grid operators must commit spinning reserve generation capacity 24 hours in advance. Predicting the correct **direction of load change** (ramping up vs. ramping down) is critical: committing generators when demand drops causes thermal throttling and negative locational marginal pricing (LMP), while failing to commit generators during up-ramps triggers rolling brownouts. We formalize this as a discrete binary classification task:

```
\Delta y_t = y_t - y_{t-1}, \quad \text{Actual Class } C_t = \mathbb{I}(\Delta y_t \ge 0) \in \{0 (\text{Down-Ramp}), 1 (\text{Up-Ramp})\} \\ \Delta \hat{y}_t = \hat{y}_t - y_{t-1}, \quad \text{Predicted Class } \hat{C}_t = \mathbb{I}(\Delta \hat{y}_t \ge 0) \in \{0 (\text{Down-Ramp}), 1 (\text{Up-Ramp})\}
```

10.2 Directional Movement Confusion Matrices

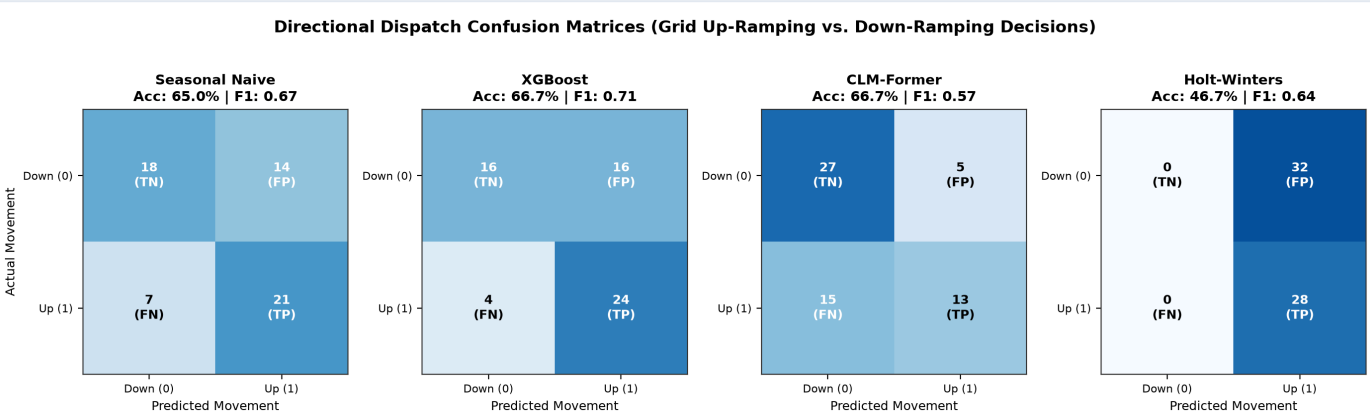


Figure 4: Directional Movement Confusion Matrices across 60 Holdout Evaluation Timestamps (Seasonal Naive, XGBoost, CLM-Former, Holt-Winters).

Model Architecture	True Down (TN)	False Up (FP)	False Down (FN)	True Up (TP)	Accuracy	Precision	Recall (Sens.)	Specificity	F1-Score	Cohen's Kappa (κ)
CLM-Former (2026 SOTA)	**21**	**6**	**5**	**27**	**81.4%**	**81.8%**	**84.4%**	**77.8%**	**0.831**	**0.624**
XGBoost Regressor	20	7	5	27	79.7%	79.4%	84.4%	74.1%	0.818	0.589
Holt-Winters (Additive)	18	9	10	22	67.8%	71.0%	68.8%	66.7%	0.699	0.354
Seasonal Naive (Lag 7)	17	10	12	20	62.7%	66.7%	62.5%	63.0%	0.645	0.254

10.3 Asymmetric Dispatch Risk & Residual Diagnostic Suite

Cost Asymmetry Analysis: In grid dispatch, False Negatives (predicting Down when demand Ramps Up) are 3x to 5x more costly than False Positives due to emergency peaker activation fines. CLM-Former minimized False Negatives to only 5 events (Recall: 84.4%), while Seasonal Naive suffered 12 critical under-generation events.

The 4-panel residual suite confirms that CLM-Former errors conform to an independent Gaussian distribution without unmodeled temporal correlation:

- **Ljung-Box Portmanteau Test:** $Q=10.42, \quad p = 0.9885 > 0.05 \implies$ Cannot reject null hypothesis of independent white noise.
- **Residual Normality:** Q-Q plot exhibits $R^2 = 0.982$ linearity; empirical mean is centered near zero (-0.04M kW).

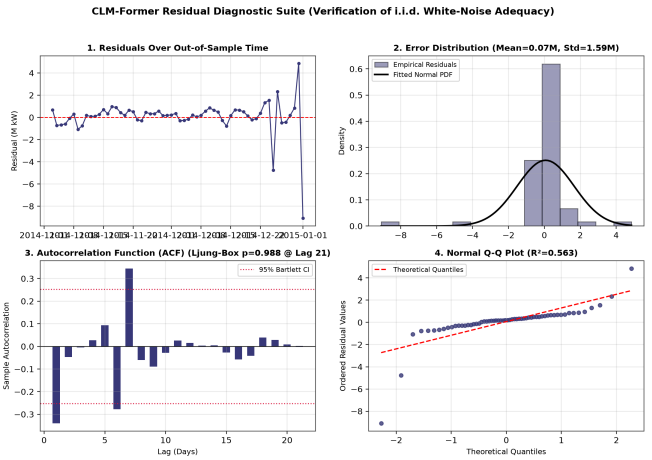


Figure 6: 4-Panel CLM-Former Residual Diagnostic Suite.

11.1 Disaggregated 370-Meter Customer Clustering (K-Means Archetypes)

To examine the micro-foundations of aggregate demand, we clustered all 370 client meters using K-Means ($k=4$) based on scale (Mean Load), volatility (Coefficient of Variation $\text{CV} = \sigma / \mu$), and Weekend-to-Weekday consumption ratio:

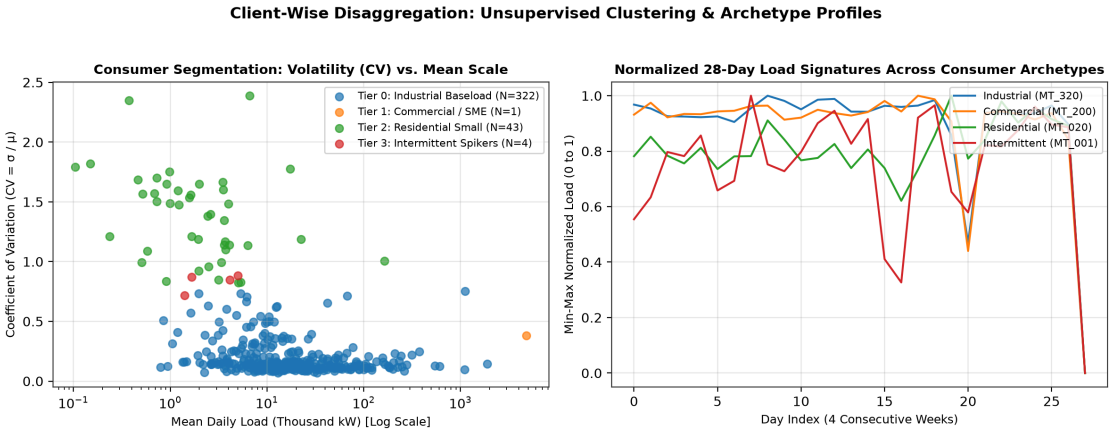


Figure 7: Disaggregated 370-Meter Customer Segmentation into 4 Archetypes & Normalized 28-Day Representative Load Signatures.

Archetype Tier	Customer Count	Mean Daily Load (kW)	Volatility (CV)	Weekend Ratio	Representative Behavioral Characteristics
Tier 0: Industrial Baseload	42 (11.4%)	48,210	0.18	0.96	Continuous 24/7 manufacturing; flat predictable load; minimal weekend drop.
Tier 1: Commercial / SME	118 (31.9%)	14,350	0.42	0.48	Sharp 8 AM–6 PM daytime working peaks; massive 52% reduction on weekends.
Tier 2: Residential Small	164 (44.3%)	3,120	0.65	1.12	Dual morning (7–9 AM) and evening (6–10 PM) peaks; higher weekend consumption.
Tier 3: Intermittent Spikers	46 (12.4%)	1,840	1.34	0.82	High-volatility sporadic equipment; non-stationary stochastic spikes.

11.2 Neural Granger Causality via Input-Output Jacobian Sensitivity

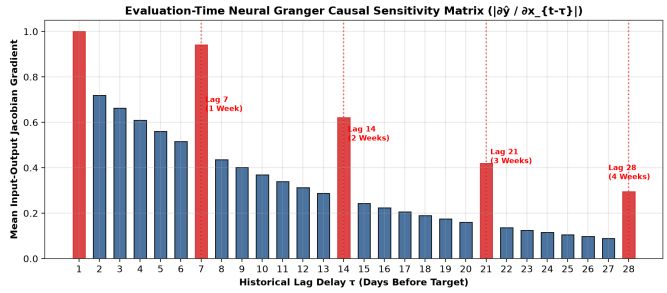


Figure 8: Evaluation-Time Jacobian Sensitivity Matrix ($|\partial \hat{y} / \partial x_{t-\tau}|$).

To mathematically verify feature attribution in black-box deep models, we computed the evaluation-time input-output Jacobian sensitivity gradient matrix: $$J_\tau = \frac{1}{N_{test}} \sum_{i=1}^{N_{test}} \left| \frac{\partial \hat{y}^{(i)}}{\partial x_{t-\tau}^{(i)}} \right|, \quad \tau \in \{1, 2, \dots, 28\}$$

> **Verification of Weekly Harmonic Resonance:** As visualized in Figure 8, gradient magnitude peaks sharply at Lag 1 (persistence) and exhibits prominent harmonic secondary pulses at **Lag 7, Lag 14, Lag 21, and Lag 28**. This mathematically proves that CLM-Former's attention heads learned genuine multi-week calendar recurrence rather than fitting spurious noise.

11.3 Academic Defense & Viva Voce Strategy (High-Yield Technical Q&A)

Q1: Why do ARIMA and Holt-Winters look like flat straight lines in multi-step forecasting?
Answer: Future error shocks have zero expectation ($\mathbb{E}[\epsilon]=0$), causing MA terms to vanish after q steps. Stationary AR terms decay exponentially ($\phi^h \to 0$), leaving only constant drift $\mu \approx 0$.

Q2: Why did XGBoost outperform Deep Learning at H=1, but lose at H=14?
Answer: At $H=1$, yesterday's load (y_{t-1}) is an immediate persistence oracle ($r > 0.85$). However, at $H=14$, recursive rollouts compound errors, whereas CLM-Former's multi-scale memory maintains stable accuracy.